23CSE202 DATABASE MANAGEMENT SYSTEMS
S3 B.Tech CSE
Department of Computer Science and Engineering
Amrita School of Computing, Amritapuri Campus

PROJECT ABSTRACT

Group No : C5
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Project Title : Smart Campus Resource Booking System

Abstract:

Campus facilities such as classrooms, laboratories, study rooms, sports venues, equipment, and parking spaces are often managed through manual registers or separate systems, leading to scheduling conflicts, low utilization, and administrative inefficiencies. The proposed Smart Campus Resource Booking System is a centralized platform that enables efficient, conflict-free reservation and management of shared campus facilities and resources through real-time availability tracking and automated scheduling.

Problems

- Double booking of classrooms, labs, and equipment.
- The absence of a centralized system makes it difficult for users to know the real-time availability of resources.
- Administrative tasks such as approvals, maintenance scheduling, and record keeping are time-consuming and prone to errors.

Objectives / Solutions Offered

- To provide a single platform for viewing, reserving, and managing all campus resources.
- To ensure fair and transparent allocation through automated conflict detection and role-based approvals.

- Automatically prevent scheduling conflicts and maintain waitlists.
- Support approvals, notifications, and utilization analytics.

Proposed Database System

- Relational database storing users, resources, reservations, maintenance records, and notifications.
- Technologies to be used: PostgreSQL(database),
  Web frontend (HTML/CSS/JavaScript),
  Backend in Python or Java.

Intended Users

- Students
- Faculty members
- Campus administrators and maintenance staff

Major Functionalities

- Secure authentication with role-based access control.
- Resource search with real-time availability display.
- Booking, modification, and cancellation of reservations.
- Automatic prevention of scheduling conflicts and support for waitlists.
- Notification services for approvals, reminders, and maintenance updates.
- Analytical dashboards and reports showing occupancy trends and resource utilization.

By digitizing and integrating the reservation process, the Smart Campus Resource Booking System improves efficiency, transparency, and user satisfaction while serving as a practical demonstration of advanced database concepts.

The system allows many users to make bookings at the same time while keeping the data accurate and consistent. It also stores past booking information and generates reports that help administrators use campus resources more effectively and plan for future needs.